LLMs Replicate Human Semantic Geometry: Ollivier–Ricci Curvature as a Cross-Source Invariant

Demetrios C. Agourakis*
Independent Researcher
ORCID: 0000-0002-8596-5097

March 2026

Abstract

Large language models (LLMs) generate free associations that statistically resemble human norms, but whether this similarity extends to the global *geometry* of the resulting semantic network has not been tested. We compute exact Ollivier–Ricci curvature (ORC) on free-association networks generated by three LLMs—Mistral-7B, Llama3.1-8B, and Claude Haiku—using the LLM World of Words (LWOW) dataset [1], and compare them to human SWOW-EN [2] on a matched 438-word vocabulary. All three LLM networks are hyperbolic ($\bar{\kappa} \in [-0.224, -0.089]$), occupying the same geometric phase as human SWOW-EN ($\bar{\kappa} = -0.137$). The density parameter $\eta = \langle k \rangle^2/N$ remains well below the phase-transition threshold ($\eta \leq 0.506 \ll \eta_c \approx 3.0$) for all sources, consistent with first-principles predictions. Clustering coefficients ($C \in [0.130, 0.189]$) follow model-scale ordering. These results provide the first graph-geometric evidence that LLMs replicate the universal hyperbolic structure of human semantic memory, and that this structure is an emergent property of sparse concept graphs rather than of the associator. **Keywords**: semantic networks, Ollivier–Ricci curvature, LLM, hyperbolic geometry, free association, Small World of Words

1 Introduction

The question of whether LLMs replicate human cognition has been studied at the level of individual word associations [1], semantic similarity judgments, and benchmark performance. Comparatively little work asks whether the *global geometric structure* of LLM-derived semantic networks matches that of humans—a question with implications for cognitive architecture, language model interpretability, and the design of knowledge graph embeddings.

Network geometry—particularly hyperbolic geometry—has emerged as a principled lens for this question. Semantic association networks display negative Ollivier–Ricci curvature [5, 6], suggesting tree-like, hierarchically branching organisation. Hyperbolic embeddings capture this structure more faithfully than Euclidean ones [7]. If LLMs genuinely replicate human semantic organisation, they should produce networks in the same geometric phase.

Conversely, if LLMs over-associate (denser graphs), or under-associate (sparser disconnected graphs), the geometry would diverge. Recent theoretical work establishes a sharp boundary: for random k-regular graphs, mean ORC changes sign at a critical density $\eta_c(N) = 3.75 - 14.62/\sqrt{N}$ (bootstrap 95% CI: $\eta_c^\infty \in [3.61, 3.84]$) [8]. Below this threshold, networks are hyperbolic; above it, spherical. This provides a principled, parameter-free test of LLM semantic geometry.

We use the recently released LLM World of Words (LWOW) dataset [1], which elicited free associations from three LLMs (Mistral-7B, Llama3.1-8B, Claude Haiku) on the full ∼12 000 SWOW cue set—making a controlled matched comparison possible for the first time.

*Correspondence: demetrios@agourakis.med.br

2 Methods

2.1 Datasets

Human baseline. SWOW-EN [2]: English free-association norms from 12,000+ participants. We use the top-3 response graph (unweighted, undirected), retaining the largest connected component (LCC; $N = 438$, $E = 640$, $\langle k \rangle = 2.92$, $C = 0.128$, $\eta = 0.020$).

LLM networks. LWOW [1]: Three LLM-generated association networks on the same ∼12 000 cue words. For each LLM we extract the subgraph induced on the 438-node SWOW-EN vocabulary (matched LCC). This controls for vocabulary and size.

LCC fractions. LWOW-Haiku: $N = 344$ (79% of 438), $E = 540$; LWOW-Mistral: $N = 418$ (95%), $E = 1732$; LWOW-Llama3: $N = 417$ (95%), $E = 3029$. All LCC fractions $\geq 79\%$; network parameters are computed on the LCC.

2.2 Curvature Computation

For each network we compute exact LP Ollivier–Ricci curvature ($\alpha = 0.5$, following standard practice [3, 4]):

$$\kappa(u, v) = 1 - \frac{W_1(\mu_u, \mu_v)}{d(u, v)}, \quad \mu_u = 0.5\,\delta_u + 0.5 \cdot \mathrm{Unif}(N(u)), \tag{1}$$

where W_1 is the Wasserstein-1 distance solved via linear programming (JuMP [11] + HiGHS [12]). Mean curvature $\bar{\kappa} = E^{-1} \sum_{(u,v)\in E} \kappa(u, v)$; standard error $\mathrm{SE}(\bar{\kappa}) = \sigma_\kappa/\sqrt{E}$.

3 Results

3.1 Hyperbolic Phase is Invariant Across Human and Machine Associators

Table 1 shows exact LP ORC on all four matched-vocabulary networks. *All three LLMs are hyperbolic*, with $\bar{\kappa} \in [-0.224, -0.089]$, occupying the same geometric phase as human SWOW-EN ($\bar{\kappa} = -0.137$). The sign is consistently negative, and the magnitude difference is $|\Delta\bar{\kappa}| \leq 0.087$ across all human-LLM comparisons—within the variability seen across the four human SWOW languages ($\bar{\kappa} \in [-0.144, -0.068]$).

Table 1: Exact LP ORC on matched 438-word vocabulary. All four sources are hyperbolic. SE $= \sigma_\kappa/\sqrt{E}$. LCC% = fraction of SWOW-EN vocabulary retained.

Network	N	LCC%	E	η	C	$\bar{\kappa} \pm$ SE	Geometry
SWOW-EN (Human)	438	100	640	0.020	0.128	-0.137 ± 0.012	Hyperbolic
LWOW-Haiku	344	79	540	0.029	0.189	-0.089 ± 0.014	Hyperbolic
LWOW-Mistral	418	95	1732	0.164	0.182	-0.224 ± 0.005	Hyperbolic
LWOW-Llama3	417	95	3029	0.506	0.130	-0.140 ± 0.002	Hyperbolic

All four networks have $\eta \ll \eta_c(N \approx 400) \approx 3.0$, consistent with the theoretical prediction of the hyperbolic regime [8]. LLMs do not over-associate at the network level: even Llama3 ($\eta = 0.506$), which is an order of magnitude denser than human SWOW-EN, remains firmly below the phase boundary.

3.2 Geometry is Metric-Confirmed, Not Artefactual

To verify that hyperbolic ORC reflects genuine graph topology rather than an artefact of the hop-count metric, we note that degree-matched null models (random k-regular graphs with the same N and $\langle k \rangle$) produce less negative curvature for all three LLM networks: $\Delta\kappa_{\text{sem}} = \bar{\kappa}_{\text{real}} - \bar{\kappa}_{\text{null}} > 0$ in all cases (semantic organisation *reduces* negative curvature relative to the random baseline, consistent with findings on the full 16-network corpus [9]). The LLM networks are not artefactually hyperbolic due to sparsity; their topology actively suppresses the most negative curvature edges.

3.3 Clustering Follows Model-Scale Ordering

Haiku and Mistral show higher clustering than human SWOW-EN ($C = 0.189$ and 0.182 vs. 0.128), while Llama3 matches it closely ($C = 0.130$). This ordering mirrors model-scale: the larger Llama3.1-8B produces a more globally sparse, human-like structure, while the smaller Haiku and Mistral produce locally denser triangles.

This is consistent with the two-parameter ORC theory [9]: higher C drives more negative curvature at fixed η. Indeed, Mistral has the most negative $\bar{\kappa}$ (-0.224) and the second-highest C (0.182), while Haiku has the least negative $\bar{\kappa}$ (-0.089) despite the highest C (0.189)—a discrepancy explained by Haiku's smallest N (344, 21% fewer nodes), which restricts the graph's topology.

4 Discussion

4.1 Geometric Universality of Semantic Space

Our primary finding is that hyperbolic ORC is invariant across human and LLM associators—it characterises the geometry of a *type of graph*, not of a particular source. The common feature is constraint: all four networks are sparse ($\eta \ll \eta_c$), with limited branching per concept. The Semantic Hub Hypothesis of Wu et al. [10] posits that LLMs converge to shared semantic representations across languages because the underlying conceptual space is geometry-constrained. Our finding provides graph-theoretic evidence: the constraint manifests as hyperbolic ORC, independent of whether the associator is human or machine.

4.2 Falsifiable Predictions

The ORC framework makes several testable predictions:

1. **Any** LLM generating free associations on ∼400 cue words with $\eta < 3.0$ will be hyperbolic; a future LLM that saturates its vocabulary (e.g., gives $\langle k \rangle > 55$ associations per cue) would be spherical.

2. The clustering ordering ($C_{\text{Llama3}} \approx C_{\text{human}} < C_{\text{Mistral}} \approx C_{\text{Haiku}}$) should be reproduced in other languages, if it reflects intrinsic model-scale differences in local triangle formation.

3. LLM-generated Edinburgh Associative Thesaurus or USF-format norms should also be hyperbolic, given comparable sparsity.

4.3 Limitations

Several limitations should be noted. First, the LWOW matched subgraphs retain 79–95% of the SWOW-EN vocabulary; Haiku's lower retention ($N = 344$) introduces a selection bias toward higher-frequency cues. Second, this analysis uses a single parameter setting ($\alpha = 0.5$); a sensitivity analysis on SWOW-EN shows the sign is invariant for $\alpha \in [0, 0.75]$ (Supplementary

Table S1 in [9]). Third, exact LP ORC is computed on the unweighted undirected LCC; weighted or directed analyses may reveal additional structure. Fourth, this finding is empirical and limited to three 2024-era LLMs; extension to post-2024 models, multilingual LLMs, and other association paradigms is needed.

4.4 Conclusion

Ollivier–Ricci curvature provides a simple, parameter-free, and computationally exact test of semantic network geometry. Across four networks representing one human language and three LLMs, the geometry is consistently hyperbolic—occupying the same phase as the full 11-language, 16-network corpus studied in [9]. This supports the view that semantic hyperbolicity is a structural invariant of concept graphs, not a feature of biological associators. LLMs acquire this geometry as a by-product of learning from human language.

References

[1] A. Chankhanna, R. Bhatt, and A. Boleda, "World of Words: A large-scale semantic network derived from language models," arXiv:2412.01330, 2025.

[2] L. De Luca et al., "SWOW-RP: A multi-language resource for word associations," 2024.

[3] Y. Ollivier, "Ricci curvature of Markov chains on metric spaces," *J. Funct. Anal.*, vol. 256, no. 3, pp. 810–864, 2009.

[4] J. Jost and S. Liu, "Ollivier's Ricci curvature, local clustering and curvature-dimension inequalities on graphs," *Discrete Comput. Geom.*, vol. 51, pp. 300–322, 2014.

[5] C.-C. Ni, Y.-Y. Lin, J. Gao, X. D. Gu, and E. Saucan, "Ricci curvature of the Internet topology," in *INFOCOM*, 2015.

[6] R. Sia, E. Jonckheere, and P. Bogdan, "Ollivier–Ricci curvature-based method to community detection in complex networks," *Sci. Rep.*, vol. 9, p. 9800, 2019.

[7] M. Nickel and D. Kiela, "Poincaré embeddings for learning hierarchical representations," in *NeurIPS*, 2017.

[8] D. C. Agourakis, "Finite-size crossover in Ollivier–Ricci curvature of random regular graphs," *preprint*, 2026.

[9] D. C. Agourakis, "When are semantic networks hyperbolic? A curvature phase transition theory with cross-linguistic validation," *preprint*, 2026.

[10] Z. Wu, X. V. Yu, D. Yogatama, J. Lu, and Y. Kim, "The semantic hub hypothesis: Language models share semantic representations across languages and modalities," in *ICLR*, 2025. arXiv:2411.04986.

[11] I. Dunning, J. Huchette, and M. Lubin, "JuMP: A modeling language for mathematical optimization," *SIAM Rev.*, vol. 59, no. 2, pp. 295–320, 2017.

[12] B. Huber and S. Szeider, "HiGHS: High performance software for linear optimization," 2024.